

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460786.6699 +/- 0.0053 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.136 +/- 0.016
Transit depth (Rp/Rs)^2	1.85 +/- 0.43 [%]
Orbital Inclination (inc)	89.85 +/- 1.23
Airmass coefficient 1 (a1)	1.3512 +/- 0.007
Airmass coefficient 2 (a2)	-0.143 +/- 0.011
Scatter in the residuals of the lightcurve fit is	0.52 %
Best Comparison Star	None
Optimal Aperture	3.88
Optimal Annulus	11.83
Transit Duration (day)	0.0967 +/- 0.0059

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.136 $\pm$ 0.016 vs 0.11 (deviation 1.63 σ)
Duration fit vs published	0.0967 d vs 0.1208 d (deviation 4.09 σ)
Mid-time O-C	-8.0 min
Depth SNR	8.5
Residual scatter	0.52 %

Light curve

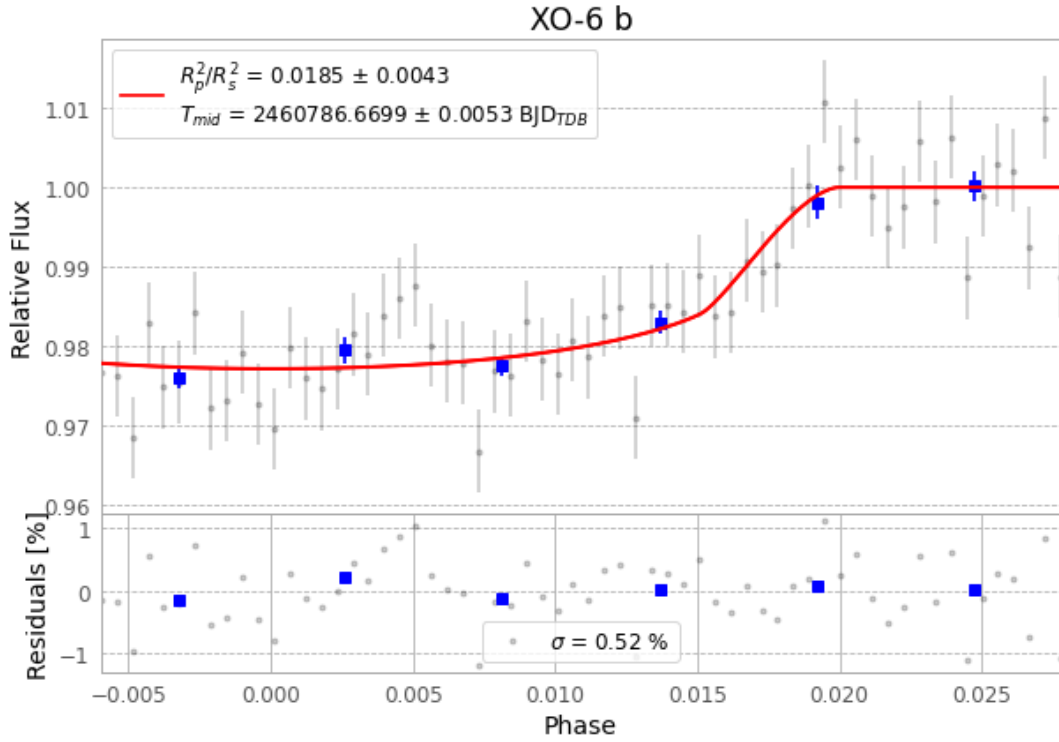


Figure 1 — FinalLightCurve_XO-6 b_2025-04-20.png (SHA-256 f6e140080ac73827...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [329, 289], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 e8acbd11cfa4ab19...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

62 FITS frames (41 MB), XO-6250421033316.FITS → XO-6250421063616.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-6-250421.log (2f6ea8cc4318...), FinalLightCurve_XO-6 b_2025-04-20.png (f6e140080ac7...), FinalLightCurve_XO-6 b_2025-04-20.csv (caaf38b7b2c1...)

Compute resources

Wall time 213.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 00:08Z.